Harmonic Oscillator Short Notes.

Formulas.

$$a_\pm = \frac{1}{\sqrt{2\hbar m\omega}}\left(m\omega x \mp ip\right).$$

$$[a_-, a_+] = 1.$$

$$H = \hbar\omega\left(a_-a_+ - \frac{1}{2}\right)$$

$$H = \hbar\omega\left(a_+a_- + \frac{1}{2}\right).$$

$$H(a_+\psi) = (E + \hbar\omega)\,\psi.$$

$$H(a_+^n\psi) = (E + n\hbar\omega)\,\psi.$$

$$H(a_-^n\psi) = (E - n\hbar\omega)\,\psi.$$

- $a_+^2\psi$ — $E + 2\hbar\omega$
- $a_+\psi$ — $E + \hbar\omega$
- ψ — E
- $a_-\psi$ — $E - \hbar\omega$
- $a_-^2\psi$ — $E - 2\hbar\omega$

Energy of $a_+\psi = E + \hbar\omega$.

a_-

$$\psi_0 = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega x^2}{2\hbar}} \qquad E_0 = \frac{\hbar\omega}{2} \qquad \boxed{a_-\psi_0 = 0}$$

zero point energy.

$$\boxed{\begin{array}{ll} \psi_n = a_+^n\left(\dfrac{m\omega}{\pi\hbar}\right)^{1/4} \cdot e^{-\frac{m\omega x^2}{2\hbar}} & E_n = \left(n + \dfrac{1}{2}\right)\hbar\omega \\ a_+\psi_n = \sqrt{n+1}\,\psi_{n+1} & a_-\psi_n = \sqrt{n}\,\psi_{n-1} \end{array}}$$

For Harmonic oscillators.

$$\langle x\rangle = 0, \quad \langle p\rangle = 0, \quad \langle x^2\rangle = \left(n + \frac{1}{2}\right)\frac{\hbar}{m\omega}$$

$$\langle p^2\rangle = \left(n + \frac{1}{2}\right)\hbar m\omega.$$

$$\underbrace{\langle V\rangle}_{PE} = \left(n + \frac{1}{2}\right)\frac{\hbar\omega}{2} = \underbrace{\langle T\rangle}_{KE}$$

$$\langle V\rangle = \langle T\rangle = \frac{E_n}{2}$$

Just like Classical mech.

$E_n \rightarrow$ const

_/_/_

Morse Potential $\rightarrow$ $V = D_e\left(1 - e^{-\beta x}\right)^2$ Not imp

Important points.

$\rightarrow$ Energy is const for an harmonic oscillator.
$E_n = \langle T\rangle + \langle V\rangle$. $\langle T\rangle = \langle V\rangle = \frac{E_n}{2}$.

$\rightarrow$ For a diatomic molecule replace $m \rightarrow \mu$ everywhere.

** $\rightarrow$ If ψ satisfies the schrödinger equation $H\psi = E\psi$, with energy E then $a_+\psi$ and $a_-\psi$ also satisfy with energy $E + \hbar\omega$, $E - \hbar\omega$.

↳ also $(a_+)^n$ & $(a_-)^n$. ↓

We can calculate energies of all levels by knowing ANY one of them

$\rightarrow$ Wave functions are either symmetric (even) or antisymmetric (odd).

$$\psi(x) = \psi(-x) \quad \text{or} \quad \psi(x) = -\psi(-x).$$

$\rightarrow$ The particle has finite probability to be found beyond the classical turning points (it penetrates the barrier)

↳ Quantum Tunneling.

$\rightarrow$ As n increases probability of finding a particle into different regions also increases and for a very large n, it is equally likely everywhere.

becomes continuous.

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→ The selection rule for an harmonic oscillator is $\Delta n = \pm 1$.
For an anharmonic oscillator is $\Delta n = \pm 1, \pm 2, \pm 3 \ldots$
intensities for $\Delta n = \pm 2, \pm 3 \ldots$ are much less than $\Delta v = \pm 1$.

→ Energy levels for harmonic oscillators are equally spaced, but for anharmonic oscillators they are not!

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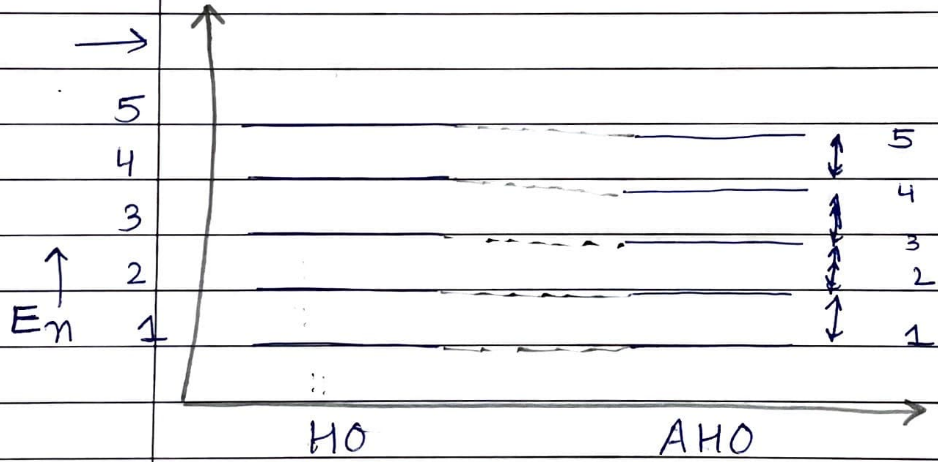

Gap keeps on decreasing between n.

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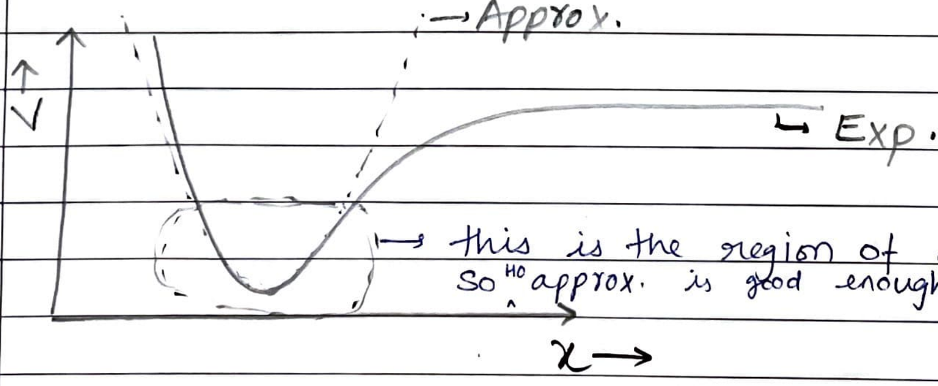

→ this is the region of our concern so HO approx. is good enough.

→ Force const. for diatomic molecules are of the order $10^2 - 10^3$ N/m.

→ Vibrational Frequency, for a diatomic molecule is of the order 10^{14} Hz.
↳ IR spectroscopy.